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Journal of Pharmacy Practice 2014 27: 46 originally published online 24 September 2013

DOI: 10.1177/0897190013504954

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Journal of Pharmacy Practice
2014, Vol. 27(1) 46-52
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DOI: 10.1177/0897190013504954
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Abstract

The primary goal of seamless care is improved patient outcomes and improved standards of care for patients with cancer. The pharmacy service of the Newfoundland Cancer Treatment and Research Foundation conducted a randomized control study that measured clinical and humanistic outcomes of a pharmacist-directed seamless care program in an ambulatory oncology clinic. This article focuses on the intervention group, particularly the identification of drug-related problems (DRPs) and utilization of health care services as well the satisfaction of 3 types of health professionals with the services provided by the pharmacist-directed seamless care program. Overall, the seamless care pharmacist (SCP) identified an average of 3.7 DRPs per intervention patient; the most common DRP reported was a patient not receiving or taking a drug therapy for which there is an indication. The SCP identified more DRPs in patients receiving adjuvant treatment compared to those receiving palliative treatment. On average, family physicians, oncology nurses, and hospital pharmacists were satisfied with the SCP intervention indicating that they agreed the information collected and distributed by the SCP was useful to them. Pharmacist-directed seamless care services in an ambulatory oncology clinic have a significant impact on clinical outcomes and processes of patient care. The presence of a SCP can help identify and resolve DRPs experienced by patients in an outpatient oncology clinic, ensuring that patients are receiving the highest standard of care.

Keywords

drug-related problems, medication reconciliation, oncology pharmacists, seamless care, pharmaceutical care

Introduction

Although the exact terminology may vary, *seamless care* is a concept that has been widely viewed as being a fundamental component in the optimal delivery of health care services. Within pharmacy, seamless care has been defined as "... the desirable continuity of care delivered to a patient in the health care system across the spectrum of caregivers and their environments. Pharmacy care is carried out without interpretation such that when one pharmacist ceases to be responsible for the patient's care, another pharmacist or health care professional accepts responsibility for the patient's care."¹ Where seamless care is provided, effort is placed on developing multidisciplinary teams that work together across any and all transitions of care.² Regrettably, in many health systems, an effective means of communication regarding a patient's drug therapy has not been established across the continuum of care. However, such communication is vital given that drugs may be added or discontinued from a patient's regimen or dosing may be altered during hospitalization. One study

reported that 71% of the patients aged 65 and older had at least one of their admission medications discontinued by the time they were discharged from hospital, accounting for 40% of all admission medications.³

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Patients, caregivers, and community pharmacists may be unclear as to what medication changes have been made in the inpatient setting and the reasons for these changes, which may subsequently lead to the patient experiencing drug-related problems (DRPs). Seamless care complements, and indeed enables, the pharmaceutical care model of pharmacy practice. The concept of pharmaceutical care requires the pharmacist to take responsibility for identifying and rectifying DRPs, providing a therapeutic monitoring plan and ensuring that patients receive the most appropriate medicine and achieve their desired quality of life.⁴ A seamless approach to health care delivery affords the infrastructure through which pharmaceutical care may be provided and a means of achieving timely and accurate communication among all members of the health care team.

Although seamless care is widely accepted in the profession of pharmacy at a conceptual level, its implementation as a standard of care has been infrequent to date. Thus, a randomized controlled trial to evaluate the impact of a pharmacist-directed seamless care service from the hospital to the community is warranted. The objective of the research was to determine the impact of pharmacist-directed seamless care services in an ambulatory oncology service on process of care and clinical outcomes. In this article, we will be discussing the impact of the seamless care pharmacist (SCP) on the intervention group, specifically the identification of DRPs and utilization of health care services as well the satisfaction of family physicians, oncology nurses, and hospital pharmacists with the services provided by the pharmacist-directed seamless care program.

Methodology

Study Design

The study was conducted at the Dr H. Bliss Murphy Cancer Centre (DHBMcC) in St John's Newfoundland under the direction of the Newfoundland Cancer Treatment and Research Foundation (NCTRF). The NCTRF is now the provincial cancer care program of Eastern Health that provides medical oncology, radiation oncology, and supportive care to patients with solid tumor cancer in Newfoundland and Labrador.

Patients considered for the study were medical patients with cancer receiving intravenous chemotherapy from any one of the regional health boards, the St John's region or in the community. The full inclusion/exclusion criterion is below under study population. Patients were randomized 1:1 to the intervention group or the control group in the clinical trials department using a random number generator.

The intervention patients received a visit from the SCP at the DHBMcC prior to the initiation of chemotherapy treatment. The SCP conducted standardized comprehensive medication history interview and medication reconciliation. Information collected on the patient's medication history was recorded on a medication history form, which was attached to the patient's chart. Upon the receipt of the intervention patient's chemotherapy order, the SCP performed a thorough

medication safety check to verify the order against established regimen protocols, including a drug interaction check, recalculation of the dose, and verification of pertinent laboratory values. All pertinent patient information was used to generate a seamless care report. Reports were created individually for each patient and summarized their visit to the medical oncology unit. It described the patient's current medications, chemotherapy regimen, common adverse events, laboratory and pathology data, antiemetics prescribed, and any other pertinent information to be transferred electronically to the other health care providers. The patient's hospital pharmacist, oncology nurse, and attending physician received copies of the report.

The SCP counseled the patient on their treatment, identified, and resolved any DRPs. A series of telephone follow-up consultations were conducted with the patient throughout the treatment to identify and manage DRPs. Patients were followed throughout their course of therapy, with consultations taking place 2 days postchemotherapy. Patients were provided contact information of the SCP and were encouraged to contact them if they had questions or concerns throughout their course of treatment.

Patients randomized to the control group did not receive pharmaceutical care but did receive a chemotherapy order check by the SCP. This was the standard of care for the majority of patients receiving chemotherapy due to the limited pharmacy resources of the cancer care program.

Study Population

For inclusion, patients had to fulfill the following criteria: at least 19 years of age, prescribed parental chemotherapy, have an Eastern Collaborative Oncology Group performance status of zero to two, and capable of completing the treatment diary and the quality of life and satisfaction questionnaires. Patients were excluded if they had symptomatic or untreated brain metastases, enrolled in other clinical trials, or were unable to give consent.

Participants were provided a written information sheet and asked to sign a consent form in accordance to the requirements of the Memorial University of Newfoundland and Labrador ethics review board.

Data Collection

Data were collected using several questionnaires and surveys aimed at both clinical and humanistic outcomes. To determine the impact of the SCP on the health care team, follow-up surveys were mailed to the family physicians, oncology nurses, and hospital pharmacists of each patient in the intervention group.

Upon enrollment, patients were provided with a diary designed by the research team. One section of the diary recorded the patients' health care service utilization. Patients were encouraged to include (1) visits to their family doctor, (2) visits to a specialist, (3) visits to the emergency room, (4) hospital admissions, (5) use of ambulance services, and (6) calls to health care workers. Patients were also asked to complete a daily grading scale to record their chemotherapy

toxicities. Diaries were collected from each patient at the end of the study period. The SCP routinely monitored the patient's care. DRPs were tabulated during SCP interactions with patients and the health care team and were categorized according to the type and severity and documented on the seamless care assessment record (Appendix A). To investigate whether there were any differences in the number and types of DRPs for patients receiving adjuvant or metastatic treatment, the intervention group was further divided into adjuvant and metastatic groups.

Data Analysis

All data for the control and intervention groups were compiled into spreadsheets using Microsoft Excel[®]. Univariate statistical analysis was done using SPSS version 15.0. Statistical significance was determined using chi-square (χ^2) tests. Significance was set at the $\alpha = .05$ level. For the purposes of this article, data from the intervention group were only used. This was done to better understand the impact of the SCP on the medical care received by the intervention group. Other articles examined the differences between the control and the intervention groups.

Results

A total of 200 patients were enrolled in the study; 100 were randomized to the control group and 100 were randomized to the intervention group. The demographic characteristics can be found in Table 1. There were no significant differences found between the control and the intervention group demographics. One patient was lost in the intervention group before the SCP intervention, and they have been excluded from the analysis of DRPs and health professional needs.

Medication History Review and Medication Reconciliation

The SCP conducted standardized comprehensive medication history interview and medication reconciliation for all patients within the intervention group. After reviewing the patient's medical history and hospital charts, the SCP identified an average of 2.57 medications per patient. The SCP performed further medication reconciliation by contacting the community pharmacies of each intervention patient, and often identified medications that were not included in the hospital charts and medication history interviews. The average number of medications obtained after interview or phone call with patient's community pharmacy by SCP grew to 3.96 per patient.

Number and Type of DRPs

By the end of the study period, the SCP identified an average of 3.7 DRPs per patient in the intervention arm (Table 2). Overall, 389 DRPs in the intervention group were identified by the SCP. There were no significant differences or correlation in DRPs by

Table 1. Demographic and Clinical Characteristics of the Intervention and Control Patients.^a

Characteristic	Control (n = 100)	Intervention (n = 100)	χ^2	P value
Gender (n)				
Male	26	35	1.644	.200
Female	74	65		
Mean age, years	55.48	58.76	52.323	.182
Proximity to cancer clinic (n)				
Urban	38	37	0.034	.854
Rural	62	62		
Did not answer	0	1		
Cancer type (n)				
Adjuvant	83	74	2.035	.154
Metastatic	17	26		
Tumor type (n)				
Lung	8	12	11.356	.414
Breast	53	44		
Prostate	1	0		
Colorectal	26	33		
Urothelial carcinoma bladder	1	1		
Renal	1	2		
Pancreatic	2	3		
Esophageal	0	2		
Stomach	6	1		
Testicular	1	1		
Cervical	0	1		
Primary unknown	1	0		

^aOne patient in the intervention dropped out.

patient age, gender, or proximity to the cancer center. However, there was a significant difference in the number of DRPs detected between patients being treated with curative intent (adjuvant patients) and patients being treated for palliation (metastatic patients). Within the intervention group, there were 74 patients in the adjuvant treatment setting and 26 in the metastatic treatment setting. The mean number of DRPs in metastatic patients (95% confidence interval) was 2.6 (1.5-3.8) compared to 4.1 (3.3-5.0) DRPs in adjuvant patients ($P = .037$; Table 2).

The most common DRP identified for both adjuvant and metastatic patients was "a patient requiring drug therapy for an indication and is not receiving/taking this therapy," which accounted for nearly 40% ($n = 148$) of all DRPs identified (Table 2). This was followed by "a patient not taking/receiving the prescribed drug appropriately" ($n = 57$, 15.2%) and "a patient taking/receiving too little drug" ($n = 39$, 10.4%).

Self-Reported Health Service Utilization

Even with the presence of a SCP the intervention group did seek out additional health care interventions (Table 3). The most common health care professional need was a visit to a family physician ($n = 41$, 41%), followed by the need to visit an emergency room ($n = 20$, 20.4%) and hospital admission ($n = 15$, 15.3%). There were 11 visits to a specialist and 11 calls to a health care worker for the intervention group (11.2%).

Table 2. Types and Numbers of Drug-Related Problems Identified by the Seamless Care Pharmacist in Both Adjuvant and Metastatic Patients Within the Intervention Group.^a

Drug-related problem identified	Adjuvant (n = 305)	Metastatic (n = 69)	Total (n = 374)	% of total
The patient is taking/receiving a drug for which there is no valid indication	0	0	0	0
The patient requires drug therapy for an indication and is not receiving/taking this therapy	117	31	148	39.6
The patient is taking/receiving the wrong drug or drug product	0	1	1	0.3
The patient is taking/receiving too little drug	31	8	39	10.4
The patient is taking/receiving too much drug	5	0	5	1.4
The patient is not taking/receiving the prescribed drug appropriately	52	4	56	15.2
The patient is experiencing an adverse drug reaction (defined as not dose related)	31	5	36	9.6
The patient is experiencing a drug–drug, drug–disease, drug–food, or drug–laboratory interaction	27	10	37	9.6
Drug interaction check	22	3	25	6.7
Compassionate access/special authority request	20	7	27	7.2

^aThe first 8 drug-related problems represent those articulated by Hepler and Strand.⁴ The bottom 2 were created specifically for this study.

Table 3. Self-Reported Health Care Service Utilization.

Health care professional needs	No.	%
Visits to family physician	41	41.8
Visits to specialist	11	11.2
Visits to emergency room	20	20.4
Hospital admissions	15	15.3
Use of ambulance services	0	0
Call to a health care worker	11	11.2

Satisfaction Surveys

Satisfaction surveys were distributed to family physicians, hospital pharmacists, and oncology nurses associated with the SCP intervention group patients (Table 4). A total of 56 surveys were completed, 16 (28.6%) by hospital pharmacists, 13 (23.2%) by family physicians, and 27 (48.2%) by oncology nurses. On average, health care professionals were satisfied with the SCP intervention indicating that they agreed the information collected and distributed by the SCP was useful to them. Although on average hospital pharmacists and family physicians found that the seamless care report assisted them in identifying and resolving DRPs (3.19 and 3.42, respectively), oncology nurse responses indicated a neutral feeling to the statement (2.33).

Discussion

This study indicates that the presence of a SCP can help to identify and resolve the number of DRPs experienced by patients in an outpatient oncology clinic. Extraordinary developments in the field of oncology are producing more complex drugs and intense regimens, which in many cases are being given outside the inpatient setting. With these medical advancements comes a greater need for specialized health care professionals to help manage the complicated issues that arise with new, more complex drug therapies. Pharmacists involved in providing pharmaceutical care services to patients with cancer work in a dynamic environment where they are required to have the knowledge, skills, and values necessary to identify drug

interactions, toxicities, adverse effects, parenteral drug stabilities, and incompatibilities as well as administration requirements or restrictions.

The SCP conducted standardized comprehensive medication history interview with all patients within the intervention group, identifying an average of 2.57 medication per patient; after an interview or phone call with patient's community pharmacy, the average medication per patient increased to 3.96. Discrepancies between what the patient reports, hospital charts, and community pharmacy files can put the patient at serious risk of dangerous drug interactions and other DRPs. These results highlight the impact an SCP can have on patient safety by conducting a complete medication reconciliation.

Overall, the SCP in this study identified an average of 3.7 DRPs per intervention patient, indicating that the pharmaceutical care practice model in an ambulatory oncology clinic can help to identify and resolve preventable drug-related morbidity. Of interest was the most common DRP reported of a patient not receiving or taking a drug therapy for which there is an indication. This particular DRP upholds the need for increased use of SCP within the oncology setting to improve communication across all oncology team members, including hospital pharmacists, physicians, and oncology nurses. It was indicated that the SCP identified more DRP in patients receiving adjuvant treatment compared to those receiving palliative treatment. This finding is not unexpected, in most cases adjuvant chemotherapy is more intense than palliative and may potentially lead to more adverse effects and more complicated drug therapy. Advancements in screening techniques has led to earlier detection of cancer, and more patients are being treated in the adjuvant setting.^{5,6} The increase in adjuvant chemotherapy leads to a growing need for pharmacists who can work closely with patients and their health care professionals to help achieve improved clinical outcomes. Our findings substantiate the importance of an SCP in the management of patients with cancer, especially in the adjuvant setting.

Buy-in from other health care professionals and key stakeholders is crucial when instituting a change in clinical protocol.

Table 4. Health Care Practitioner Satisfaction With the SCP Intervention.^a

Question	Hospital pharmacists (n = 16)	Family physicians (n = 13)	Oncology nurses (n = 27)
Having access to the following patient information is useful to my practice			
(1) Medication history and relevant medical history	3.88	3.92	3.30
(2) The appropriate dose and cycle frequency for the regimen	3.69	3.62	3.37
(3) The administration guidelines	3.81	N/A	3.11
(4) A link to the chemotherapy protocol	3.75	3.62	3.11
The information found on the seamless care report assisted me in			
(1) Identifying and resolving drug related problem(s)?	3.19	3.42	2.33
Overall, having access to the information found on the seamless care report			
(1) Enhanced efficiencies in service provision (ie, saved time)	3.56	3.46	2.59
(2) Facilitated the provision of pharmaceutical care	3.73	N/A	N/A

Abbreviations: N/A, not applicable, SCP, seamless care pharmacist.

^aBased on a 5-point Likert-type Scale with a range from strongly disagree (zero) to strongly agree (four).

The results of this study indicate that other health care team members found the presence of an SCP useful in helping them to perform their jobs; these data provide supportive evidence that an SCP would be a welcomed addition in other clinics and settings. Having a designated health care professional in a coordinating role greatly enhances each team member's ability to respond to the needs of the patient and to communicate potential DRPs as they occur.

There are some limitations to this study. First, this intervention included only 1 SCP at 1 ambulatory oncology site in Atlantic Canada. As such, these results may not be generalizable to other oncology clinics. Nonetheless, it is expected that similar results would be generated in other clinics due to the positive effect that the SCP had on communication and patient care. Second, this article reports the outcomes of the intervention group only. More insight is expected to be obtained through a comparison of the intervention and control groups, including differences in quality of life measures and health care professional needs.

Conclusions

Pharmacist-directed seamless care services in an ambulatory oncology clinic have a significant impact on clinical and humanistic outcomes and processes of patient care. The presence of an SCP can help identify and resolve DRPs experienced by patients in an outpatient oncology clinic, ensuring that patients receive the most appropriate medicine and enhancing their quality of life. Furthermore, an SCP is regarded as an asset to other health care professionals within the patient's circle of care. Medical advancements in the field of oncology are achieving more complex drugs and regimens, creating a pressing need for specialized health care professionals, like pharmacists, which can optimize safety and effectiveness of the treatment and improve outcomes. Our findings substantiate the importance of an SCP in the management of patients with cancer, especially in the adjuvant setting. This study supports the implementation of seamless care as a standard of care within the outpatient oncology clinic and beyond.

Appendix A

Seamless Care Assessment Record

Patient Name: _____

Study Number: _____

Toxicity	Grade	Intervention Made	Call Back Date	DRP Category
Nausea				
Vomiting				
Anorexia				
Diarrhea				
Constipation				
Stomatitis				
Fever				
Infection				
Febrile Neutropenia				
Alopecia				
Rash				
Pain				
Sensory				
Motor				
Fatigue				
Dyspnea				
Taste Changes				
Other				

COMMENTS:

Interviewer: _____

Date completed: _____

DRP categories

1. Patient is taking/receiving a drug for which there is no valid indication
2. The patient requires drug therapy for an indication and is not receiving/taking this therapy
3. The patient is not taking/receiving the appropriate drug or drug product
4. The patient is taking/receiving too little drug
5. The patient is taking/receiving too much drug
6. The patient is not taking/receiving the prescribed drugs appropriately
7. The patient is experiencing an adverse drug reaction
8. The patient is experiencing a drug-drug, drug-food, or drug-laboratory interaction

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship and/or publication of this article: Funded through unrestricted research grants from Pfizer, Amgen, and Roche.

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